Homework 4 Solutions

Section 2.4

② e

$$\left[\begin{array}{ccc|ccc}3&5&0&1&0&0\\3&7&1&0&1&0\\1&2&1&0&0&1\end{array}\right]\xrightarrow[R_1\leftrightarrow R_3]{}\left[\begin{array}{ccc|ccc}1&2&1&0&0&1\\3&7&1&0&1&0\\3&5&0&1&0&0\end{array}\right]\xrightarrow[\substack{R_2-2R_1\\R_3-3R_1}]{}\left[\begin{array}{ccc|ccc}1&2&1&0&0&1\\0&1&-2&0&1&-3\\0&-1&-3&1&0&-3\end{array}\right]\xrightarrow[R_3+R_2]{}\left[\begin{array}{ccc|ccc}1&2&1&0&0&1\\0&1&-2&0&1&-3\\0&0&-5&1&1&-6\end{array}\right]\xrightarrow[-\frac{1}{5}R_3]{}\left[\begin{array}{ccc|ccc}1&2&1&0&0&1\\0&1&-2&0&1&-3\\0&0&1&-\frac{1}{5}&-\frac{1}{5}&\frac{6}{5}\end{array}\right]$$

$$\xrightarrow[\substack{R_1-R_3\\R_2+2R_3}]{}\left[\begin{array}{ccc|ccc}1&2&0&\frac{1}{5}&\frac{1}{5}&-\frac{1}{5}\\0&1&0&-\frac{2}{5}&\frac{3}{5}&-\frac{3}{5}\\0&0&1&-\frac{1}{5}&-\frac{1}{5}&\frac{6}{5}\end{array}\right]\xrightarrow[R_1-2R_2]{}\left[\underbrace{\begin{matrix}1&0&0\\0&1&0\\0&0&1\end{matrix}}_{I_3}\middle|\begin{matrix}1&-1&1\\-\frac{2}{5}&-\frac{2}{5}&-\frac{2}{5}\\-\frac{1}{5}&-\frac{1}{5}&\frac{6}{5}\end{matrix}\right]$$

So the inverse is

$$\boxed{\begin{bmatrix}3&5&0\\3&7&1\\1&2&1\end{bmatrix}^{-1}=\frac{1}{5}\begin{bmatrix}5&-5&5\\-2&-2&-2\\-1&-1&6\end{bmatrix}}$$

i

$$\left[\begin{array}{ccc|ccc}3&1&2&1&0&0\\1&-1&3&0&1&0\\1&2&4&0&0&1\end{array}\right]\xrightarrow[R_1\leftrightarrow R_2]{}\left[\begin{array}{ccc|ccc}1&-1&3&0&1&0\\3&1&2&1&0&0\\1&2&4&0&0&1\end{array}\right]\xrightarrow[\substack{R_2-3R_1\\R_3-R_1}]{}\left[\begin{array}{ccc|ccc}1&-1&3&0&1&0\\0&4&-7&1&-3&0\\0&3&1&0&-1&1\end{array}\right]\xrightarrow[R_2-R_3]{}\left[\begin{array}{ccc|ccc}1&-1&3&0&1&0\\0&1&-8&1&-2&-1\\0&3&1&0&-1&1\end{array}\right]\xrightarrow[R_3-3R_2]{}\left[\begin{array}{ccc|ccc}1&-1&3&0&1&0\\0&1&-8&1&-2&-1\\0&0&25&-3&5&4\end{array}\right]$$

$$\xrightarrow[\frac{1}{25}R_3]{}\left[\begin{array}{ccc|ccc}1&-1&3&0&1&0\\0&1&-8&1&-2&-1\\0&0&1&-\frac{3}{25}&\frac{1}{5}&\frac{4}{25}\end{array}\right]\xrightarrow[\substack{R_2+8R_3\\R_1-3R_3}]{}\left[\begin{array}{ccc|ccc}1&-1&0&\frac{9}{25}&\frac{2}{5}&-\frac{12}{25}\\0&1&0&\frac{1}{25}&-\frac{2}{5}&\frac{7}{25}\\0&0&1&-\frac{3}{25}&\frac{1}{5}&\frac{4}{25}\end{array}\right]\xrightarrow[R_1+R_2]{}\left[\underbrace{\begin{matrix}1&0&0\\0&1&0\\0&0&1\end{matrix}}_{=I_3}\middle|\begin{matrix}\frac{2}{5}&0&-\frac{1}{5}\\\frac{1}{25}&-\frac{2}{5}&\frac{7}{25}\\-\frac{3}{25}&\frac{1}{5}&\frac{4}{25}\end{matrix}\right]$$

So the inverse is

$$\boxed{\begin{bmatrix}3&1&2\\1&-1&3\\1&2&4\end{bmatrix}^{-1}=\frac{1}{25}\begin{bmatrix}10&0&-5\\1&10&7\\-3&-5&4\end{bmatrix}}$$

3 d

The matrix form of this system is $A\vec{x}=\vec{b}$, where

$$A=\begin{bmatrix}1&4&2\\2&3&3\\4&1&4\end{bmatrix}\quad\text{and}\quad\vec{b}=\begin{bmatrix}1\\-1\\0\end{bmatrix}$$

We find A^{-1} first

$$\left[\begin{array}{ccc|ccc}1&4&2&1&0&0\\2&3&3&0&1&0\\4&1&4&0&0&1\end{array}\right]\xrightarrow[\substack{R_2-2R_1\\R_4-4R_1}]{}\left[\begin{array}{ccc|ccc}1&4&2&1&0&0\\0&-5&-1&-2&1&0\\0&-15&-4&-4&0&1\end{array}\right]\xrightarrow[R_3-3R_2]{}\left[\begin{array}{ccc|ccc}1&4&2&1&0&0\\0&-5&-1&-2&1&0\\0&0&-1&2&-3&1\end{array}\right]\xrightarrow[-R_3]{}\left[\begin{array}{ccc|ccc}1&4&2&1&0&0\\0&-5&-1&-2&1&0\\0&0&1&-2&3&-1\end{array}\right]\xrightarrow[\substack{R_2+R_3\\R_1-2R_3}]{}\left[\begin{array}{ccc|ccc}1&4&0&5&-6&2\\0&-5&0&-4&4&-1\\0&0&1&-2&3&-1\end{array}\right]$$

$$\xrightarrow[-\frac{1}{5}R_2]{}\left[\begin{array}{ccc|ccc}1&4&0&5&-6&2\\0&1&0&\frac{4}{5}&-\frac{4}{5}&\frac{1}{5}\\0&0&1&-2&3&-1\end{array}\right]\xrightarrow[R_1-4R_2]{}\left[\underset{\overset{\shortparallel}{I_3}}{\begin{matrix}1&0&0\\0&1&0\\0&0&1\end{matrix}}\middle|\begin{matrix}\frac{9}{5}&-\frac{14}{5}&\frac{6}{5}\\\frac{4}{5}&-\frac{4}{5}&\frac{1}{5}\\-2&3&-1\end{matrix}\right],\ \text{So } A \text{ is invertible and } A^{-1}=\frac{1}{5}\begin{bmatrix}9&-14&6\\4&-4&1\\-10&15&-5\end{bmatrix}$$

As A is invertible, the system $A\vec{x}=\vec{b}$ has exactly one solution:

$$\vec{x}=A^{-1}\vec{b}=\frac{1}{5}\begin{bmatrix}9&-14&6\\4&-4&1\\-10&15&-5\end{bmatrix}\begin{bmatrix}1\\-1\\0\end{bmatrix}=\boxed{\frac{1}{5}\begin{bmatrix}23\\8\\-25\end{bmatrix}} \qquad \left(\text{i.e. } x=\frac{23}{5},\ y=\frac{8}{5},\ z=-5\right)$$

⑤ $\boxed{d}$

$(I-2A^T)^{-1}=\begin{bmatrix}2&1\\1&1\end{bmatrix} \Rightarrow I-2A^T=\begin{bmatrix}2&1\\1&1\end{bmatrix}^{-1}$ (if this inverse exists):

$$\left[\begin{array}{cc|cc}2&1&1&0\\1&1&0&1\end{array}\right]\xrightarrow{R_1-R_2}\left[\begin{array}{cc|cc}1&0&1&-1\\1&1&0&1\end{array}\right]\xrightarrow{R_2-R_1}\left[\begin{array}{cc|cc}1&0&1&-1\\0&1&-1&2\end{array}\right]$$

with the left block $=I_2$, so $\begin{bmatrix}2&1\\1&1\end{bmatrix}$ is invertible with inverse $\begin{bmatrix}1&-1\\-1&2\end{bmatrix}$.

$$\Rightarrow I-2A^T=\begin{bmatrix}1&-1\\-1&2\end{bmatrix}\Rightarrow -2A^T=\begin{bmatrix}1&-1\\-1&2\end{bmatrix}-I=\begin{bmatrix}0&-1\\-1&1\end{bmatrix}\Rightarrow A^T=\begin{bmatrix}0&\frac{1}{2}\\\frac{1}{2}&-\frac{1}{2}\end{bmatrix}\Rightarrow\boxed{A=\begin{bmatrix}0&\frac{1}{2}\\\frac{1}{2}&-\frac{1}{2}\end{bmatrix}}$$

$\boxed{f}$

$$\left(\begin{bmatrix}1&0\\2&1\end{bmatrix}A\right)^{-1}=\begin{bmatrix}1&0\\2&2\end{bmatrix}\Rightarrow A^{-1}\begin{bmatrix}1&0\\2&1\end{bmatrix}^{-1}=\begin{bmatrix}1&0\\2&2\end{bmatrix}\Rightarrow A^{-1}=\begin{bmatrix}1&0\\2&2\end{bmatrix}\begin{bmatrix}1&0\\2&1\end{bmatrix}=\begin{bmatrix}1&0\\6&2\end{bmatrix}\Rightarrow A=\begin{bmatrix}1&0\\6&2\end{bmatrix}^{-1} \quad (\text{If it exists})$$

$$\left[\begin{array}{cc|cc}1&0&1&0\\6&2&0&1\end{array}\right]\xrightarrow{R_2-6R_1}\left[\begin{array}{cc|cc}1&0&1&0\\0&2&-6&1\end{array}\right]\xrightarrow{\frac{1}{2}R_2}\left[\begin{array}{cc|cc}1&0&1&0\\0&1&-3&\frac{1}{2}\end{array}\right],\quad \text{so}\quad \boxed{A=\begin{bmatrix}1&0\\-3&\frac{1}{2}\end{bmatrix}}$$

(left block $=I_2$)

$\boxed{h}$

$$(A^{-1}-2I)^T=-2\begin{bmatrix}1&1\\1&0\end{bmatrix}\Rightarrow A^{-1}-2I=-2\begin{bmatrix}1&1\\1&0\end{bmatrix}=\begin{bmatrix}-2&-2\\-2&0\end{bmatrix}\Rightarrow A^{-1}=\begin{bmatrix}-2&-2\\-2&0\end{bmatrix}+2I=\begin{bmatrix}0&-2\\-2&2\end{bmatrix}\Rightarrow A=\begin{bmatrix}0&-2\\-2&2\end{bmatrix}^{-1} \quad (\text{If it exists})$$

$$\left[\begin{array}{cc|cc}0&-2&1&0\\-2&2&0&1\end{array}\right]\xrightarrow{R_1\leftrightarrow R_2}\left[\begin{array}{cc|cc}-2&2&0&1\\0&-2&1&0\end{array}\right]\xrightarrow{R_1+R_2}\left[\begin{array}{cc|cc}-2&0&1&1\\0&-2&1&6\end{array}\right]\xrightarrow[-\frac{1}{2}R_2]{-\frac{1}{2}R_1}\left[\begin{array}{cc|cc}1&0&-\frac{1}{2}&-\frac{1}{2}\\0&1&-\frac{1}{2}&0\end{array}\right],\quad \text{so}\quad \boxed{A=-\frac{1}{2}\begin{bmatrix}1&1\\1&0\end{bmatrix}}$$

(left block $=I_2$)

⑦ We write

$$\vec{x}=\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix},\quad \vec{y}=\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix},\quad \vec{z}=\begin{bmatrix}z_1\\z_2\\z_3\end{bmatrix},\quad A=\begin{bmatrix}3&-1&2\\1&0&4\\2&1&0\end{bmatrix},\quad B=\begin{bmatrix}1&-1&1\\2&-3&0\\-1&1&-2\end{bmatrix}$$

We then have $\vec{x}=A\vec{y}$ and $\vec{z}=B\vec{y}$. If B were invertible, we could continue as follows: $\vec{y}=B^{-1}\vec{z}\Rightarrow\vec{x}=A\vec{y}=AB^{-1}\vec{z}$.
Let's check if B has an inverse:

$$\left[\begin{array}{ccc|ccc}1&-1&1&1&0&0\\2&-3&0&0&1&0\\-1&1&-2&0&0&1\end{array}\right]\xrightarrow[R_3+R_1]{R_2-2R_1}\left[\begin{array}{ccc|ccc}1&-1&1&1&0&0\\0&-1&-2&-2&1&0\\0&0&-1&1&0&1\end{array}\right]\xrightarrow[-R_3]{-R_2}\left[\begin{array}{ccc|ccc}1&-1&1&1&0&0\\0&1&2&2&-1&0\\0&0&1&-1&0&-1\end{array}\right]\xrightarrow[R_2-2R_3]{R_1-R_3}\left[\begin{array}{ccc|ccc}1&-1&0&2&0&1\\0&1&0&4&-1&2\\0&0&1&-1&0&-1\end{array}\right]\xrightarrow{R_1+R_2}\left[\begin{array}{ccc|ccc}1&0&0&6&-1&3\\0&1&0&4&-1&2\\0&0&1&-1&0&-1\end{array}\right]$$

(left block $=I_3$)

Therefore, B is invertible with inverse $B^{-1}=\begin{bmatrix}6&-1&3\\4&-1&2\\-1&0&-1\end{bmatrix}$. Thus, we obtain

$$\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}3&-1&2\\1&0&4\\2&1&0\end{bmatrix}\begin{bmatrix}6&-1&3\\4&-1&2\\-1&0&-1\end{bmatrix}\begin{bmatrix}z_1\\z_2\\z_3\end{bmatrix}=\boxed{\begin{bmatrix}12&-2&5\\2&-1&-1\\16&-3&8\end{bmatrix}\begin{bmatrix}z_1\\z_2\\z_3\end{bmatrix}}$$

(17)

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 2 & -1 & 2 & 0 & 1 & 0 \\ 0 & 2 & c & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_2-2R_1]{} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 2 & c & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3-2R_2]{} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & c & 4 & -2 & 1 \end{array}\right] \xrightarrow[\frac{1}{c}R_3]{} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 4/c & -2/c & 1/c \end{array}\right]$$

$$\xrightarrow[R_1-R_3]{} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1-4/c & 2/c & -1/c \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 4/c & -2/c & 1/c \end{array}\right] \xrightarrow[R_1+R_2]{} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1-4/c & 2/c+1 & -1/c \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 4/c & -2/c & 1/c \end{array}\right]$$

$= I_3$

So the inverse is

$$\boxed{\begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 2 \\ 0 & 2 & c \end{bmatrix}^{-1} = \frac{1}{c}\begin{bmatrix} -4-c & 2+c & -1 \\ -2c & c & 0 \\ 4 & -2 & 1 \end{bmatrix}}$$

Section 2.5

(2) a

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 3 & -1 & 0 & 1 \end{array}\right] \xrightarrow[R_2-R_1]{} \left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & -2 & -1 & 1 \end{array}\right], \text{ so } B = E_{R_2-R_1}A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} A \Rightarrow \boxed{E = E_{R_2-R_1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}}$$

c

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{array}\right] \xrightarrow[R_1\leftrightarrow R_2]{} \left[\begin{array}{cc|cc} -1 & 2 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{array}\right], \text{ so } B = E_{R_1\leftrightarrow R_2}A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} A \Rightarrow \boxed{E = E_{R_1\leftrightarrow R_2} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}$$

e

$$\left[\begin{array}{cc|cc} -1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{array}\right] \xrightarrow[-R_2]{} \left[\begin{array}{cc|cc} -1 & 1 & 1 & 0 \\ -1 & 1 & 0 & -1 \end{array}\right], \text{ so } B = E_{-R_2}A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} A \Rightarrow \boxed{E = E_{-R_2} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}}$$

3a

$$\underbrace{\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ -1 & 1 & 0 & 1 \end{array}\right]}_{= A} \xrightarrow[R_1\leftrightarrow R_2]{} \left[\begin{array}{cc|cc} -1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{array}\right] \xrightarrow[R_2-2R_1]{} \underbrace{\left[\begin{array}{cc|cc} -1 & 1 & 0 & 1 \\ 2 & 1 & 1 & -2 \end{array}\right]}_{= C}.$$

Therefore, $C = E_{R_2-2R_1}E_{R_1\leftrightarrow R_2}A$, so we may take

$$\boxed{E_1 = E_{R_1\leftrightarrow R_2} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad E_2 = E_{R_2-2R_1} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}}$$

(6) b

$$\underbrace{\left[\begin{array}{ccc|cc} 1 & 2 & 1 & 1 & 0 \\ 5 & 12 & -1 & 0 & 1 \end{array}\right]}_{= A} \xrightarrow[R_2-5R_1]{} \left[\begin{array}{ccc|cc} 1 & 2 & 1 & 1 & 0 \\ 0 & 2 & -6 & -5 & 1 \end{array}\right] \xrightarrow[R_1-R_2]{} \left[\begin{array}{ccc|cc} 1 & 0 & 7 & 6 & -1 \\ 0 & 2 & -6 & -5 & 1 \end{array}\right] \xrightarrow[\frac{1}{2}R_2]{} \left[\begin{array}{ccc|cc} 1 & 0 & 7 & 6 & -1 \\ 0 & 1 & -3 & -5/2 & 1/2 \end{array}\right].$$

(the left block $= R$, the right block $= U$)

We get $R = UA$ for $\boxed{U = \frac{1}{2}\begin{bmatrix} 6 & -1 \\ -5 & 1 \end{bmatrix} = E_{\frac{1}{2}R_2}E_{R_1-R_2}E_{R_2-5R_1}}$

Note: There are more possibilities for U.

c

$$\left[\begin{array}{cccc|ccc} 1 & 2 & -1 & 0 & 1 & 0 & 0 \\ 3 & 1 & 1 & 2 & 0 & 1 & 0 \\ 1 & -3 & 3 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3-R_1]{R_2-3R_1} \left[\begin{array}{cccc|ccc} 1 & 2 & -1 & 0 & 1 & 0 & 0 \\ 0 & -5 & 4 & 2 & -3 & 1 & 0 \\ 0 & -5 & 4 & 2 & -1 & 0 & 1 \end{array}\right] \xrightarrow{R_3-R_2} \left[\begin{array}{cccc|ccc} 1 & 2 & -1 & 0 & 1 & 0 & 0 \\ 0 & -5 & 4 & 2 & -3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & -1 & 1 \end{array}\right] \xrightarrow{-\frac{1}{5}R_2} \left[\begin{array}{cccc|ccc} 1 & 2 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -4/5 & -2/5 & 3/5 & -1/5 & 0 \\ 0 & 0 & 0 & 0 & 2 & -1 & 1 \end{array}\right]$$

(the first 3×4 block $= A$)

$$\xrightarrow{R_1-2R_2} \left[\begin{array}{cccc|ccc} 1 & 0 & 3/5 & 4/5 & -1/5 & 2/5 & 0 \\ 0 & 1 & -4/5 & -2/5 & 3/5 & -1/5 & 0 \\ 0 & 0 & 0 & 0 & 2 & -1 & 1 \end{array}\right]$$

(left block $= R$, right block $= U$)

So $R = UA$ for $\boxed{U = \frac{1}{5}\begin{bmatrix} -1 & 2 & 0 \\ 3 & -1 & 0 \\ 10 & -5 & 5 \end{bmatrix} = E_{R_1-2R_2} E_{-\frac{1}{5}R_2} E_{R_3-R_2} E_{R_3-R_1} E_{R_2-3R_1}}$

Note: There are more possibilities for U.

8 **b**

$$\left[\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 2 & 3 & 1 & 0 \end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{array}\right] \xrightarrow{R_1+2R_2} \left[\begin{array}{cc|cc} 1 & 0 & 2 & -3 \\ 0 & -1 & 1 & -2 \end{array}\right] \xrightarrow{-R_2} \left[\begin{array}{cc|cc} 1 & 0 & 2 & -3 \\ 0 & 1 & -1 & 2 \end{array}\right]$$

(first left block $= A$, last left block $= I$)

So $A^{-1} = \begin{bmatrix} 2 & -3 \\ -1 & -2 \end{bmatrix} = E_{-R_2} E_{R_1+2R_2} E_{R_2-2R_1} E_{R_1 \leftrightarrow R_2} \Rightarrow A = \left(E_{-R_2} E_{R_1+2R_2} E_{R_2-2R_1} E_{R_1 \leftrightarrow R_2}\right)^{-1}$

$= \boxed{E_{R_1 \leftrightarrow R_2} E_{R_2+2R_1} E_{R_1-2R_2} E_{-R_2}}$

Note: There is more than one solution

d

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ -2 & 2 & 15 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3+2R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & -3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 2 & 9 & 2 & 0 & 1 \end{array}\right] \xrightarrow{R_3-2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & -2 & 1 \end{array}\right] \xrightarrow[R_2-4R_3]{R_1+3R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7 & -6 & 3 \\ 0 & 1 & 0 & -8 & 9 & -4 \\ 0 & 0 & 1 & 2 & -2 & 1 \end{array}\right]$$

(first left block $= A$, last left block $= I$)

So $A^{-1} = \begin{bmatrix} 7 & -6 & 3 \\ -8 & 9 & -4 \\ 2 & -2 & 1 \end{bmatrix} = E_{R_2-4R_3} E_{R_1+3R_3} E_{R_3-2R_2} E_{R_3+2R_1} \Rightarrow A = \left(E_{R_2-4R_3} E_{R_1+3R_3} E_{R_3-2R_2} E_{R_3+2R_1}\right)^{-1}$

$= \boxed{E_{R_3-2R_1} E_{R_3+2R_2} E_{R_1-3R_3} E_{R_2+4R_3}}$

Note: There is more than one solution